Question 4: Install and configure Git on Ubuntu Linux, verify the installation, and set up global Git configuration (user name and email).

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-1-

1. System and Git installation

installation and verification

```
meenakshi@ubuntu:~$ lsb_release -d
Description:    Ubuntu 24.04.3 LTS
meenakshi@ubuntu:~$ sudo apt update && sudo apt install -y git
meenakshi@ubuntu:~$ git --version
git version 2.43.0
```

2. Global Git configuration

configuration commands

```
git config --global user.name "Meenakshi"
git config --global user.email "meenakshi16rp@gmail.com"
git config --global init.defaultBranch main
git config --global credential.helper store
git config --global core.editor nano
```

verification

```
meenakshi@ubuntu:~$ git config --global --list
user.name=Meenakshi
user.email=meenakshi16rp@gmail.com
init.defaultbranch=main
core.editor=nano
credential.helper=store
```

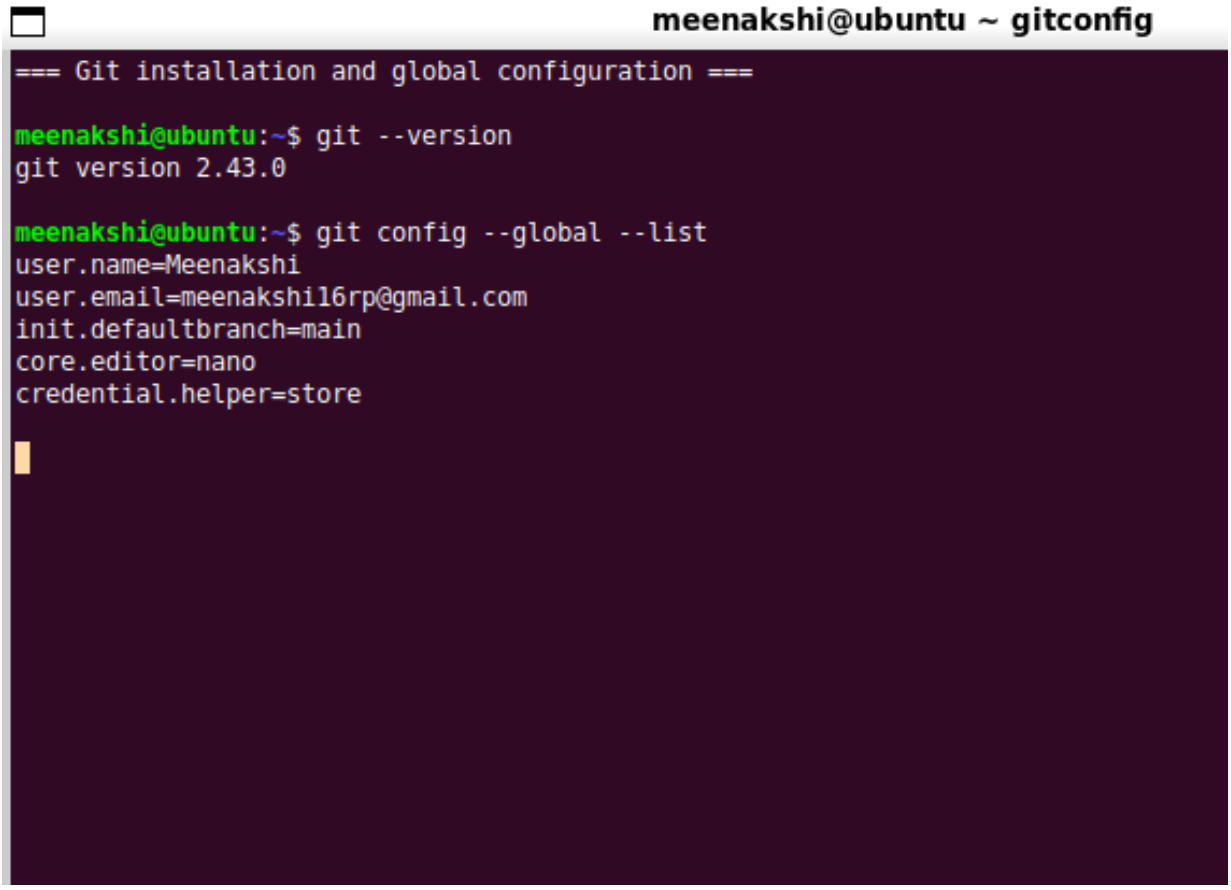


Figure: Ubuntu terminal: meenakshi@ubuntu prompt, Ubuntu 24.04.3, Git 2.43.0 and the global configuration

3. Jenkins installation

Jenkins LTS installation (systemd service on port 8080)

```
sudo apt install -y openjdk-21-jdk
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key -o \
      /usr/share/keyrings/jenkins-keyring.asc
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" \
      | sudo tee /etc/apt/sources.list.d/jenkins.list
sudo apt update && sudo apt install -y jenkins
sudo systemctl enable --now jenkins
sudo systemctl is-active jenkins
active
jenkins --version
2.580.2
```

4. Jenkins first-run configuration

The setup wizard was completed in the browser at http://localhost:8080: unlock with the initial admin password, install the suggested plugins (Pipeline, Git, GitHub Branch Source, Credentials Binding, ...), and create the administrator account.

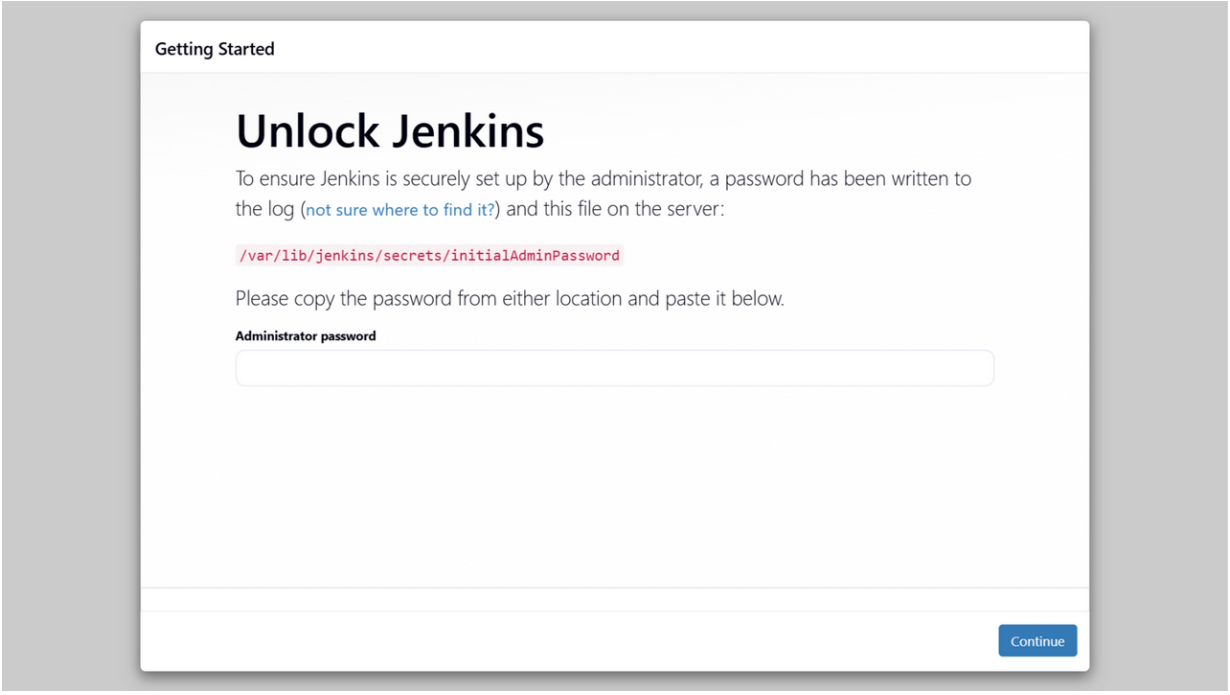


Figure: Unlock Jenkins - initial admin password from /var/lib/jenkins/secrets/initialAdminPassword

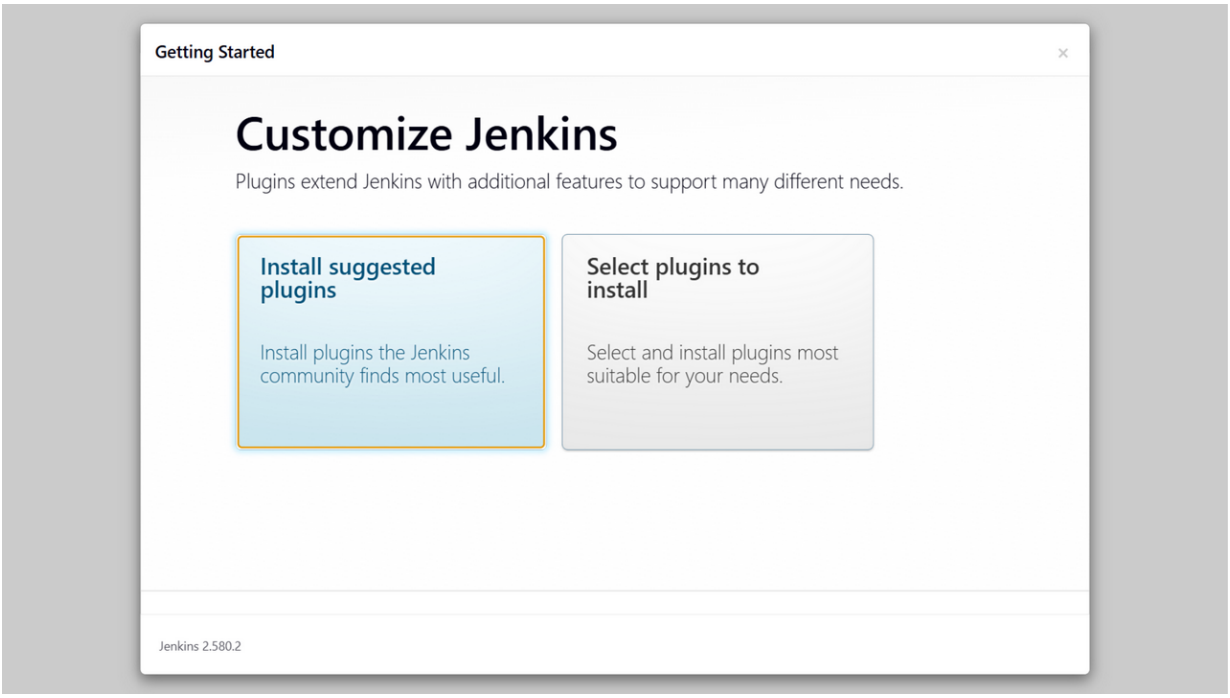


Figure: Customize Jenkins - install suggested plugins

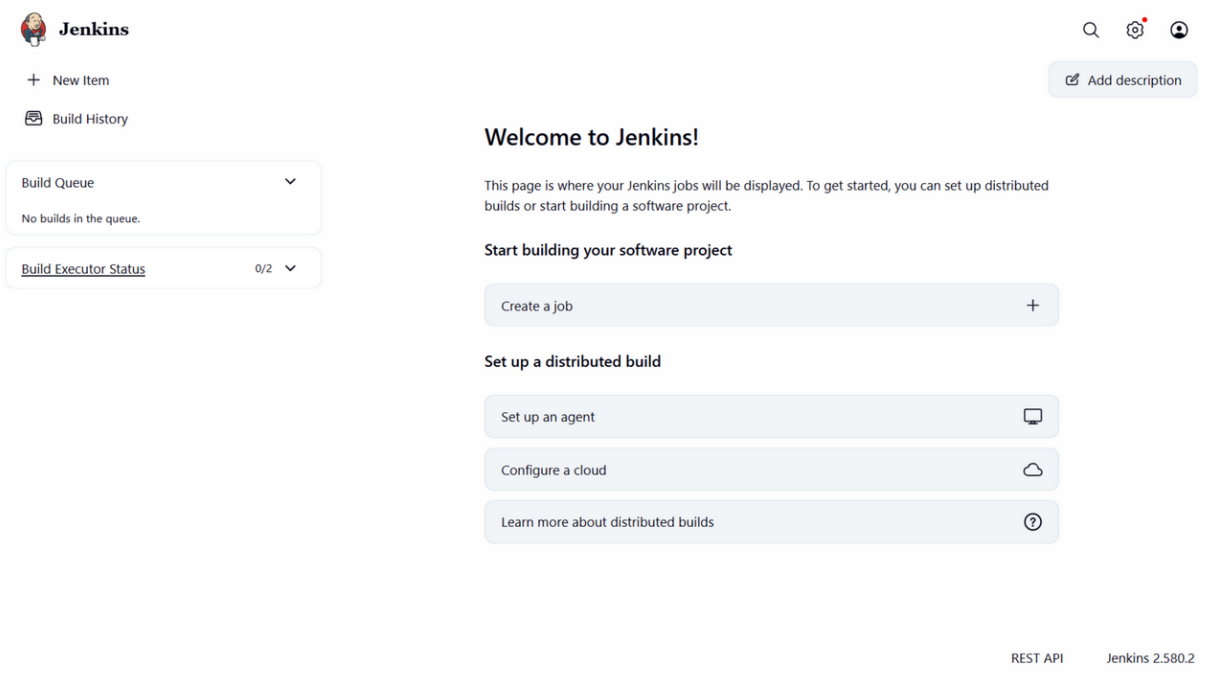


Figure: Jenkins dashboard ready (admin user: meenakshi)

5. Observations

- Jenkins runs as a systemd service (user jenkins, JAVA 21) and survived WSL restarts.
- The initial admin password file is removed after unlocking; plugins install directly from the official update center.
- Global Git identity ensures every commit made in the lab is attributed to Meenakshi.